

Fundamentals of Machine Learning

A Core Summary of Paradigms, Concepts, and Metrics

1. What is Machine Learning?

Machine Learning (ML) is a subfield of artificial intelligence focused on building systems that learn from data, identify patterns, and make decisions with minimal human intervention. Rather than hardcoding explicit rules, ML algorithms use statistical methods to optimize performance dynamically over time.

The Core Objective: The primary goal of an ML model is to *generalize* well to new, unseen datasets, avoiding both underfitting (oversimplification) and overfitting (memorization of noise).

2. Primary Paradigms of Learning

Machine learning problems are broadly categorized into three core paradigms based on the nature of the learning signal:

- **Supervised Learning:** The model trains on labeled data consisting of input-output pairs $(\mathbf{x}, y)$. The goal is to learn a mapping function $f(\mathbf{x}) = y$.
 - *Regression:* Predicting continuous numerical values (e.g., housing prices).
 - *Classification:* Predicting discrete categorical classes (e.g., distinguishing spam from legitimate email).
- **Unsupervised Learning:** The model encounters unlabeled training data $(\mathbf{x})$ and must uncover hidden structures or relationships autonomously.
 - *Clustering:* Grouping data points by geometric or statistical proximity (e.g., K-Means).
 - *Dimensionality Reduction:* Simplifying the feature space while retaining crucial variance (e.g., Principal Component Analysis).
- **Reinforcement Learning (RL):** An autonomous agent learns to maximize cumulative rewards by interacting dynamically with an environment through continuous trial and error.

3. The Bias-Variance Tradeoff

Achieving structural balance in predictive modeling requires managing two distinct sources of error:

Bias: Error caused by overly simplistic assumptions in the learning algorithm. High bias leads to **underfitting**, where the model fails to capture the underlying trend of the data.

Variance: Error arising from extreme sensitivity to small fluctuations in the training dataset. High variance leads to **overfitting**, where the model performs exceptionally on training data but fails to generalize to test data.

$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

4. Core Evaluation Metrics

Choosing the correct validation metric prevents false conclusions regarding model efficacy. The selection depends fully on the task type:

Classification Metrics

For discrete predictions, metrics are derived from the *Confusion Matrix* comprising True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN).

Metric	Mathematical Representation	Primary Use Case Target
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Symmetric datasets with balanced class frequencies.
Precision	$TP / (TP + FP)$	Minimizing false positives (e.g., spam filter precision).
Recall (Sensitivity)	$TP / (TP + FN)$	Minimizing false negatives (e.g., medical diagnoses).
F1-Score	$2 \times (Precision \times Recall) / (Precision + Recall)$	Imbalanced classes requiring a balanced harmonic mean.

Regression Metrics

For predicting continuous values, models measure distances between actual targets y_i and predictions $\hat{y}_i$:

- **Mean Absolute Error (MAE):** Simple average of absolute error distances; robust against outliers.
- **Mean Squared Error (MSE):** Heavily penalizes larger errors due to squaring operations.

5. Key Regularization Techniques

To curb overfitting and control high variance, regularization penalties are added directly to the loss function:

- **L1 Regularization (Lasso):** Adds an absolute value penalty proportional to the coefficients. It enforces feature sparsity, driving unhelpful weights completely to zero.
- **L2 Regularization (Ridge):** Adds a squared magnitude penalty to the coefficients. It diffuses weights evenly, keeping values small but non-zero.